

Power Systems

BRANCH: ELECTRICAL (EE)

COURSE CODE: EE301

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Power Systems (EE301) under the Electrical (EE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Generation & Economics

Thermal, Hydro, Nuclear, Renewable energy power plants. Load curve, load factor, diversity factor, tariff structures.

Unit 2: Transmission Line Parameters

Inductance and Capacitance calculations of single and three-phase lines. GMD and GMR calculations. Skin effect, Proximity effect, Corona discharge.

Unit 3: Performance of Lines & Insulators

Short, Medium, Long transmission lines representation. Nominal-T and Nominal-Pi models. Efficiency and regulation. Overhead insulators: Pin, Suspension, Strain types.

Unit 4: Fault Analysis & Load Flow

Symmetrical faults, unsymmetrical faults (LG, LL, LLG). Symmetrical components. Load flow analysis: Gauss-Seidel, Newton-Raphson methods.

Load Factor = Average Load / Peak Load. Symmetrical components transformation matrix.

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



